Assignment 2: Localization and Motion Planning

Due: 23:59, Oct. 26, 2026 (Monday) | 4 questions | 100 marks

If you have any questions about this assignment or its grading, please contact **Gang WANG** at `gwang2@mae.cuhk.edu.hk`.

Guidelines:

- Write your solutions with clear steps on the answer sheet, blank paper, or single-lined paper.
- Only handwritten work produced with a physical or digital pen is accepted.
- Write your full name, student ID, and page number on every answer sheet.
- Scan all answer sheets clearly, save them as one **single PDF file**, and upload it to Blackboard.
- Late submissions will **not** be considered.
- Only PDF files will be accepted.
- Keep the original hard copy after submission in case it is needed for checking.
- You may use a mobile scanning app such as CamScanner or Microsoft Lens.

QUESTION 1: Markov localization in a one-dimensional corridor — 25 MARKS

A robot moves along a one-dimensional corridor divided into ten cells, numbered $x \in \{0, 1, \ldots, 9\}$. Its initial belief assigns probability 0.25 to each of cells 0–3 and zero to every other cell. Conditional on any possible initial cell, a rightward command u_1 moves it one cell with probability 0.4 or two cells with probability 0.6:

$$p(x_0 + 1 \mid x_0, u_1) = 0.4, \qquad p(x_0 + 2 \mid x_0, u_1) = 0.6.$$

All other transition probabilities are zero. All transitions from the support of the initial belief remain within the corridor; there is no wrap-around. A subsequent observation z_1 has the following likelihoods:

x_1	0	1	2	3	4	5	6–9
$p(z_1 \mid x_1)$	0.1	0.1	0.1	0.8	0.8	0.2	0.1

The final entry applies separately to each of cells 6, 7, 8, and 9.

(a) Belief and update equations [5 marks]. Write the initial belief as a ten-entry vector in cell order 0–9. State the discrete prediction equation and the normalized measurement-update equation. Use $\overline{bel}(x_1)$ for the prediction.

(b) Prediction [10 marks]. Compute $\overline{bel}(x_1)$ for all ten cells. Show how the nonzero entries are obtained and verify that they sum to one.

(c) Measurement update [10 marks]. Compute the unnormalized weights, the normalization factor η, and the posterior $bel(x_1)$ for all ten cells. Verify its normalization.

Q1: a). Initial Distribution: $bel(X_0) = [0.25, 0.25, 0.25, 0.25, 0, 0, 0, 0]$

$\overline{bel}(x_1) = \sum_{x_0=0}^{9} P(x_1|x_0, u_1)\, bel(x_1)$ $\quad bel(x_1) = \eta P(z_1|x_1)\overline{bel}(x_1)$. $\eta = \frac{1}{\sum_{j=0}^{9} P(z_1|x_1)\overline{bel}(j)}$

b). $P(x_1=1) = P(x_0=0)P(u_1=1) = 0.25\times 0.4 = 0.1$ $\quad P(x_1=2) = P(x_0=0)P(u_1=2) + P(x_0=1)$

$P(u_1=1) = 0.25\times 0.6 + 0.25\times 0.4 = 0.25$ $\quad P(x_1=3) = P(x_0=1)P(u_1=2) + P(x_0=2)P(u_1=1) = 0.25\times$

$0.6 + 0.25\times 0.4 = 0.25$ $\quad P(x_1=4) = P(x_0=2)P(u_1=2) + P(x_0=3)P(u_1=1) = 0.25$ $\quad P(x_1=5) = P(x_0=3)$

$P(u_1=2) = 0.15$. $\quad \overline{bel}(x_1) = [0, 0.1, 0.25, 0.25, 0.25, 0.15, 0, 0, 0]$

$0.1 + 0.25\times 3 + 0.15 = 1$

c). $w(0) = 0$ $\quad w(1) = 0.1\times 0.1 = 0.01$ $\quad w(2) = 0.1\times 0.25 = 0.025$ $\quad w(3) = 0.8\times 0.25$

$= 0.2$ $\quad w(4) = 0.8\times 0.25 = 0.2$ $\quad w(5) = 0.2\times 0.15 = 0.03$ $\quad w(6) = w(7) = w(8) = w(9) = 0$.

$\sum_j w(j) = 0.465$. $\quad \eta = \frac{1}{\sum_j w(j)} = \frac{200}{93} \rightarrow bel(1) = \eta w(1) = \frac{2}{93}$, $bel(2) = \eta w(2) = \frac{5}{93}$

$bel(3) = \eta w(3) = \frac{40}{93}$ $\quad bel(4) = \eta w(4) = \frac{40}{93}$ $\quad bel(5) = \eta w(5) = \frac{2}{31}$ $\quad bel(0) = bel(6) =$

$bel(7) = bel(8) = bel(9) = 0$. $\sum_{x_1=0}^{9} bel(x_1) = 1$

QUESTION 2: One-dimensional Kalman filter localization — 25 MARKS

A robot's one-dimensional position follows the linear-Gaussian model

$$x_k = x_{k-1} + u_k + w_{k-1}, \qquad z_k = x_k + v_k,$$

where $w_{k-1} \sim \mathcal{N}(0, Q)$ and $v_k \sim \mathcal{N}(0, R)$. The noise sequences are independent across time, mutually independent, and independent of the initial state. Thus $A = B = H = 1$.

The initial posterior and noise variances are

$$\hat{x}_{0|0} = 0\,\text{m}, \quad P_{0|0} = 0.40\,\text{m}^2, \quad Q = 0.10\,\text{m}^2, \quad R = 0.25\,\text{m}^2.$$

The controls and measurements are

Step k	Control u_k (m)	Measurement z_k (m)
1	1.20	1.00
2	0.80	2.30

Use $\hat{x}_{k|k-1}, P_{k|k-1}$ for prediction and $\hat{x}_{k|k}, P_{k|k}$ for the posterior. Retain exact fractions or sufficient precision in intermediate calculations; report decimal results to at least four decimal places.

(a) Scalar filter equations [5 marks]. Write the equations for the predicted mean and variance, innovation, Kalman gain, corrected mean, and corrected variance.

(b) First step [10 marks]. Compute $\hat{x}_{1|0}, P_{1|0}$, innovation $\tilde{y}_1$, gain K_1, and posterior $\hat{x}_{1|1}, P_{1|1}$. Show substitutions.

(c) Second step [10 marks]. Starting from the posterior in (b), compute the corresponding six quantities at $k = 2$. Show substitutions.

Q2:

$x_k = x_{k-1} + u_k + w_{k-1}$ $z_k = x_k + v_k$ $A=B=H=1$. $\hat{x}_{0|0} = 0m$ $P_{0|0} = 0.4$ $Q=0.1$ $R=0.25$

a). $\hat{x}_{k|k-1} = \hat{x}_{k-1|k-1} + u_k$ $P_{k|k-1} = P_{k-1|k-1} + Q$ $\tilde{y}_k = z_k - \hat{x}_{k|k-1}$ $K_k = \frac{P_{k|k-1}}{P_{k|k-1}+R}$.

$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_k$ $P_{k|k} = (1-K_k) P_{k|k-1}$

b). $u_1 = 1.2$ $z_1 = 1$. $\hat{x}_{1|0} = 0 + 1.20 = 1.2$ $P_{1|0} = 0.4 + 0.1 = 0.5$ $\tilde{y}_1 = 1.0 - 1.2 = -0.2$

$K_1 = \frac{0.5}{0.5+0.25} = \frac{2}{3}$ $\hat{x}_{1|1} = 1.2 + \frac{2}{3} \times (-0.2) = \frac{16}{15}$ $P_{1|1} = (1-\frac{2}{3}) \times 0.5 = \frac{1}{6}$

c) $u_2 = 0.8$ $z_2 = 2.3$. $\hat{x}_{2|1} = \frac{16}{15} + 0.8 = \frac{28}{15}$ $P_{2|1} = \frac{1}{6} + 0.1 = \frac{4}{15}$ $\tilde{y}_2 = 2.3 - \frac{28}{15} = \frac{13}{30}$

$K_2 = \frac{\frac{4}{15}}{\frac{4}{15}+0.25} = \frac{16}{31}$ $\hat{x}_{2|2} = \frac{28}{15} + \frac{16}{31} \times \frac{13}{30} = \frac{324}{155}$ $P_{2|2} = (1-\frac{16}{31}) \frac{4}{15} = \frac{4}{31}$

QUESTION 3: Markov localization on a two-dimensional grid — 25 MARKS

A robot moves on a 4×4 grid. A cell is identified by (x, y), with $x, y \in \{0, 1, 2, 3\}$. The x-axis increases to the right and the y-axis increases upward. The robot's heading is known and fixed; only its position is estimated.

The initial belief is

$$bel_0(0,0) = 0.20, \qquad bel_0(1,0) = 0.60, \qquad bel_0(2,0) = 0.20,$$

and is zero at every other cell. For the command u_1, the displacement is independent of the initial cell and has the distribution

Δx	Δy	Probability
0	1	0.50
1	2	0.50

All other displacements have probability zero. All possible transitions from the nonzero initial cells remain within the grid; there is no wrap-around.

After the motion, an observation z_1 has the following likelihood $L(x, y) = p(z_1 \mid x, y)$:

$y \backslash x$	0	1	2	3
3	0	0	0	0
2	0	0.25	0.25	0
1	0	0.25	0.25	0
0	0	0	0	0

For all grid answers, show columns in order $x = 0, 1, 2, 3$ and rows from top to bottom in order $y = 3, 2, 1, 0$. Use $\overline{bel}_1(x, y)$ for the predicted belief and $bel_1(x, y)$ for the posterior.

(a) Initial belief and update equations [5 marks]. Write the initial belief as a 4×4 grid. State the discrete two-dimensional prediction equation and the normalized measurement-update equation.

(b) Two-dimensional prediction [10 marks]. Compute $\overline{bel}_1(x, y)$ for every cell. Show the contributions from the initial cells and the two possible displacements, present the complete predicted grid, and verify that its entries sum to one.

(c) Measurement update [10 marks]. Multiply the prediction by $L(x, y)$ to obtain the unnormalized grid. Calculate the normalization factor η, present the complete posterior grid, and verify its normalization.

Q3:

a)

y \ x	0	1	2	3
3	0	0	0	0
2	0	0	0	0
1	0	0	0	0
0	0.2	0.6	0.2	0

$$\overline{bel}_1(x,y) = \sum_{x_0=0}^{3} \sum_{y_0=0}^{3} P(x,y \mid x_0, y_0, u_1)\, bel_0(x_0, y_0)$$

$$bel_1(x,y) = \eta L(x,y)\, \overline{bel}_1(x,y).$$

$$\eta = \frac{1}{\sum L(x,y)\, \overline{bel}_1(x,y)}$$

b). $(\Delta x=0, \Delta y=1) = 0.5$ $(\Delta x=1, \Delta y=2) = 0.5$. $\overline{bel}_1(0,1) = P(0,0) \times P(\Delta x=0, \Delta y=1)$

$= 0.2 \times 0.5 = 0.1$ $\overline{bel}_1(1,2) = 0.2 \times 0.5 = 0.1$. $\overline{bel}_1(1,1) = 0.6 \times 0.5 = 0.3$ $\overline{bel}_1(2,2)$

$= 0.6 \times 0.5 = 0.3$ $\overline{bel}_1(2,1) = 0.2 \times 0.5 = 0.1$ $\overline{bel}_1(3,2) = 0.2 \times 0.5 = 0.1$

y \ x	0	1	2	3
3	0	0	0	0
2	0	0.1	0.3	0.1
1	0.1	0.3	0.1	0
0	0	0	0	0

$0.1 + 0.1 + 0.3 + 0.3 + 0.1 = 1.$

c). $w(1,1) = 0.25 \times 0.3 = 0.075$ $w(1,2) = 0.25 \times 0.1 = 0.025$

$w(2,1) = 0.25 \times 0.1 = 0.025$ $w(2,2) = 0.25 \times 0.3 = 0.075.$

$\eta = \frac{1}{\sum w(x)} = 5$ $bel_1(1,1) = bel_1(2,2) = 0.375.$ $bel_1(1,2) = bel_1(2,1) = 0.125$

QUESTION 4: Local navigation with a simplified dynamic window — 25 MARKS

Use the lecture's simplified planar DWA example: a holonomic point robot is controlled by Cartesian velocity (v_x, v_y). Initially,

$$(x_0, y_0) = (0, 0)\,\mathrm{m}, \qquad (v_{x,c}, v_{y,c}) = (0.50, 0)\,\mathrm{m/s}.$$

The acceleration limit applies separately to each velocity component:

$$|a_x|, |a_y| \le 0.40\,\mathrm{m/s^2}, \qquad \sqrt{v_x^2 + v_y^2} \le 1.00\,\mathrm{m/s}.$$

The control interval is $\Delta t_c = 0.50\,\mathrm{s}$, the prediction horizon is $T = 2.00\,\mathrm{s}$, and the simulation step is $\Delta t_s = 0.50\,\mathrm{s}$. Build the reachable velocity window using Δt_c. Consider the following velocity samples:

Candidate	v_x (m/s)	v_y (m/s)
A	0.60	0
B	0.60	0.20
C	0.80	0
D	0.40	−0.20

Assume every window-valid candidate has already passed collision and braking-safety checks. For prediction only, use the constant-velocity approximation from time zero:

$$x_{j+1} = x_j + v_x \Delta t_s, \qquad y_{j+1} = y_j + v_y \Delta t_s.$$

This approximation does not model the acceleration transient.

(a) Dynamic window [8 marks]. Compute the reachable interval for each velocity component using the acceleration limits and control interval. Write the intersection with the maximum-speed constraint.

(b) Candidate filtering [7 marks]. Test all four candidates against this window and state the reason for every rejection.

(c) Trajectory prediction [10 marks]. For each valid candidate, calculate the positions at $t = 0.50, 1.00, 1.50, 2.00\,\mathrm{s}$. Show the calculation method and present the results in a table.

End of assignment. Total: 100 marks.

Q4. $(x_0, y_0) = (0,0)$, $(v_{x,c}, v_{y,c}) = (0.5, 0)$, $|a_x|, |a_y| \le 0.40\ m/s^2$, $\sqrt{v_x^2 + v_y^2} \le 1.00\ m/s$.

$\Delta t_c = 0.5s$, $T = 2.00s$, $\Delta t_s = 0.5s$. a). $\Delta v_{max} = a_{max} \Delta t_c = 0.2 m/s$. $v_x \in [v_{x,c} - 0.2, v_{x,c} + 0.2] = [0.3, 0.7]$ $v_y \in [-0.2, 0.2]$

$$V = \begin{cases} 0.3 \le v_x \le 0.7 \\ (v_x, v_y): -0.2 \le v_y \le 0.2 \\ v_x^2 + v_y^2 \le 1.00 \end{cases}$$

b). A: $v_x = 0.6 \in V$, $v_y = 0 \in V$, $\sqrt{v_x^2 + v_y^2} = 0.6 \in V$: valid

B: $v_x = 0.6 \in V$, $v_y = 0.2 \in V$, $\sqrt{v_x^2 + v_y^2} = \sqrt{0.4} \in V$: valid.

C: $v_x = 0.8 > 0.7$, invalid D: $v_x = 0.4 \in V$, $v_y = -0.2 \in V$, $\sqrt{v_x^2 + v_y^2} = \sqrt{0.2} \in V$: valid.

c). $\begin{cases} x_{j+1} = x_j + v_x \Delta t_s \\ y_{j+1} = y_j + v_y \Delta t_s. \end{cases}$ Point numbers: $N = \frac{T}{\Delta t_s} = \frac{2}{0.5} = 4$.

A: $(x(t), y(t)) = (0.6t, 0)$. B: $(x(t), y(t)) = (0.6t, 0.2t)$ D: $(x(t), y(t)) = (0.4t, -0.2t)$.

	A	B	D
0.5	(0.3, 0)	(0.3, 0.1)	(0.2, -0.1)
1.0	(0.6, 0)	(0.6, 0.2)	(0.4, -0.2)
1.5	(0.9, 0)	(0.9, 0.3)	(0.6, -0.3)
2.0	(1.2, 0)	(1.2, 0.4)	(0.8, -0.4)

The Chinese University of Hong Kong
Academic Honesty Declaration Statement

Submission Details

Student Name	SUN Hejia (s1155262875)		
Year and Term	2026-2027 Term 1		
Course	ROSE-5730-- Autonomous Mobile Robots		
Assignment Marker	Dr. HAN Dongkun		
Submitted File Name	ROSE5730_Assignment2_SunHejia_1155262875.pdf		
Submission Type	Individual		
Assignment Number	2	**Due Date (provided by student)**	2026-10-26
Submission Reference Number	4664455	**Submission Time**	2026-10-22 22:42:40

Agreement and Declaration on Student's Work Submitted to VeriGuide

VeriGuide is intended to help the University to assure that works submitted by students as part of course requirement are original, and that students receive the proper recognition and grades for doing so. The student, in submitting his/her work ("this Work") to VeriGuide, warrants that he/she is the lawful owner of the copyright of this Work. The student hereby grants a worldwide irrevocable non-exclusive perpetual licence in respect of the copyright in this Work to the University. The University will use this Work for the following purposes.

(a) Checking that this Work is original

The University needs to establish with reasonable confidence that this Work is original, before this Work can be marked or graded. For this purpose, VeriGuide will produce comparison reports showing any apparent similarities between this Work and other works, in order to provide data for teachers to decide, in the context of the particular subjects, course and assignment. In addition, the Work may be investigated by AI content detection software to determine originality. However, any such reports that show the author's identity will only be made available to teachers, administrators and relevant committees in the University with a legitimate responsibility for marking, grading, examining, degree and other awards, quality assurance, and where necessary, for student discipline.

(b) Anonymous archive for reference in checking that future works submitted by other students of the University are original

The University will store this Work anonymously in an archive, to serve as one of the bases for comparison with future works submitted by other students of the University, in order to establish that the latter are original. For this purpose, every effort will be made to ensure this Work will be stored in a manner that would not reveal the author's identity, and that in exhibiting any comparison with other work, only relevant sentences/ parts of this Work with apparent similarities will be cited. In order to help the University to achieve anonymity, this Work submitted should not contain any reference to the student's name or identity except in designated places on the front page of this Work (which will allow this information to be removed before archival).

(c) Research and statistical reports

The University will also use the material for research on the methodology of textual comparisons and evaluations, on teaching and learning, and for the compilation of statistical reports. For this purpose, only the anonymously archived material will be used, so that student identity is not revealed.

I confirm that the above submission details are correct. I am submitting the assignment for:

[X] an individual project.

I have read the above and in submitting this Work fully agree to all the terms. I declare that: (i) the assignment here submitted is original except for source material explicitly acknowledged; (ii) the piece of work, or a part of the piece of work has not been submitted for more than one purpose (e.g. to satisfy the requirements in two different courses) without declaration; and (iii) the submitted soft copy with details listed in the <Submission Details> is identical to the hard copy(ies), if any, which has(have) been / is(are) going to be submitted. I also acknowledge that I am aware of the University's policy and regulations on honesty in academic work, and of the disciplinary guidelines and procedures applicable to breaches of such policy and regulations, as contained in the University website http://www.cuhk.edu.hk/policy/academichonesty/.

I declare that I have not distributed/ shared/ copied any teaching materials without the consent of the course teacher(s) to gain unfair academic advantage in the assignment/ course.

I declare that I have read and understood the University's policy on the use of AI for academic work. I confirm that I have complied with the instructions given by my teacher regarding the use of AI tools for this assignment and consent to the use of AI content detection software to review my submission.

I also understand that assignments without a properly signed declaration by the student concerned will not be graded by the teacher(s)



Signature (SUN Hejia, s1155262875)

2026/10/22.

Date

Instruction for Submitting Hard Copy / Soft Copy of the Assignment

This signed declaration statement should be attached to the hard copy assignment or submission to the course teacher, according to the instructions as stipulated by the course teacher. If you are required to submit your assignment in soft copy only, please print out a copy of this signed declaration statement and hand it in separately to your course teacher.